



مركز الشيخ مشعل الأحمد الجابر الصباح للشباب والتكامل السيادي الوطني

# HHMYC

DESIGN BASIS REGISTER

REV 1.0 · SEPTEMBER 2026

A PLEDGE AND A PROMISE... FOR THE FUTURE  
OF KUWAIT

عهد ووعد... لكويت المستقبل

STATUS: GOVERNING REGISTER

Concept stage — all values subject to survey and geotechnical investigation

Press → or swipe to continue





A ring of ten identical modules, inspired by the Kuwaiti emblem, united in symmetry to symbolise youth, unity, and vision. Each module opens toward the inner courtyard, creating a seamless flow between culture, learning, and innovation.

Sheikh Mishal Al-Ahmad Al-Jaber Al-Sabah Center for Youth and National Sovereign Integration. A coastal ring of ten identical units around an unbuilt oasis — architecture as national emblem, not a landmark object.

10

Identical units

36°

Rotational increment

46,000 m<sup>2</sup>

Total GFA

Ø 68 m

Open oasis



مركز الشيخ مشعل الأحمد الجابر الصباح  
للشباب والتكامل السيادي الوطني

# HHMYC

## DESIGN BASIS REGISTER

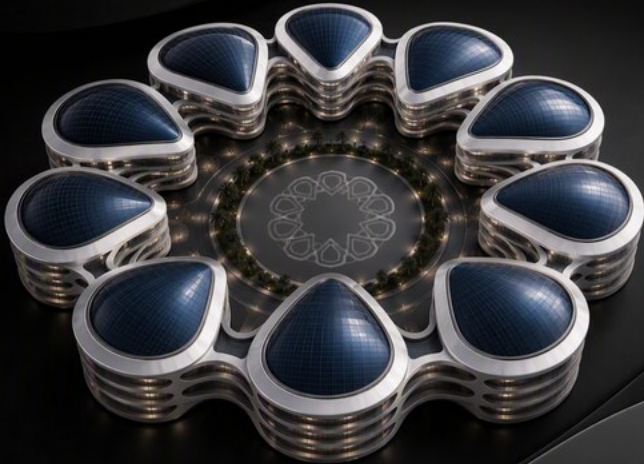
Rev 1.0 | September 2026

Sheikh Mishal Al-Ahmad Al-Jaber Al-Sabah Center  
for Youth and National Sovereign Integration (HHMC)

*A Pledge and a Promise... For the Future of Kuwait.*

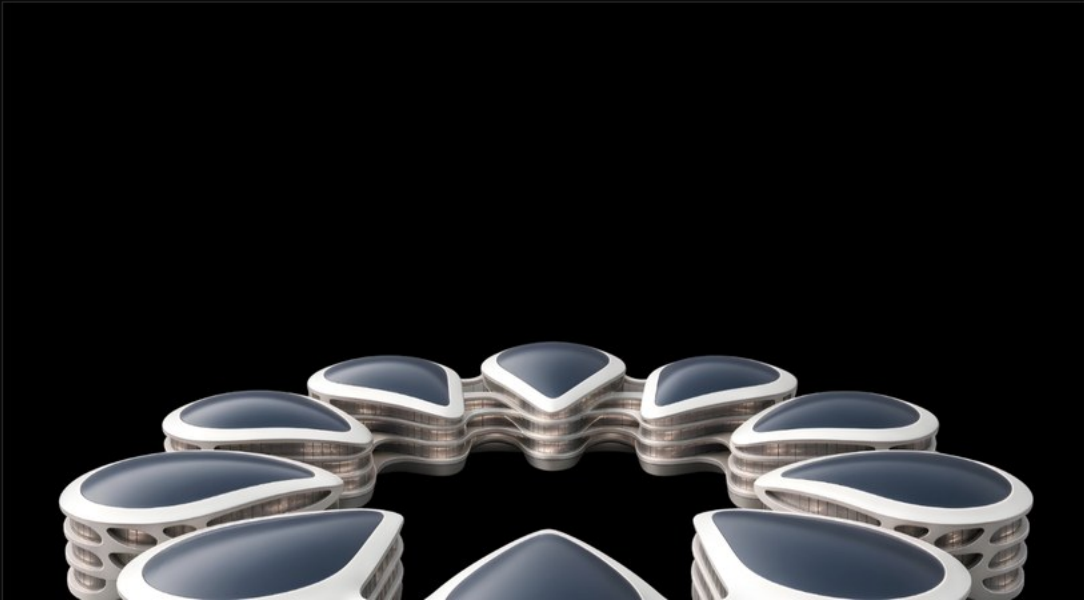
### STATUS: GOVERNING REGISTER

Every project document (strategy, catalogue, RFQ, QTO, BoO, master plan, program, control package, models) shall cite this register and shall not restate a value that differs from it. Where a document conflicts, this register prevails.



Symmetric soft-edged geometric form. Not a teardrop, not a petal.

|                             |                                                                       |
|-----------------------------|-----------------------------------------------------------------------|
| Identical units / increment | 10 / 36°                                                              |
| Unit outline                | Degree-8 Bézier B <sub>0</sub> –B <sub>n</sub> , mirrored about U = 0 |
| Unit plan L × W             | 45.760 × 32.261 m                                                     |
| Floor area / floors         | 1,050 m² × 3 levels                                                   |
| Finished floor levels       | 0.000 / +4.800 / +9.200                                               |
| Roof datum / apex           | +13.600 m / +20.400 m                                                 |
| Inner tip / oasis           | 34.000 m / Ø 68.000 m                                                 |
| Outer crown / ring          | 79.760 m / Ø 159.520 m                                                |
| Roof gridshell              | 38.0 × 28.0 m, rise 6.800 m                                           |
| Diagrid cell                | 2.4 – 3.0 m                                                           |
| Connectors                  | 10 per level × 3 levels · 40 m² each                                  |

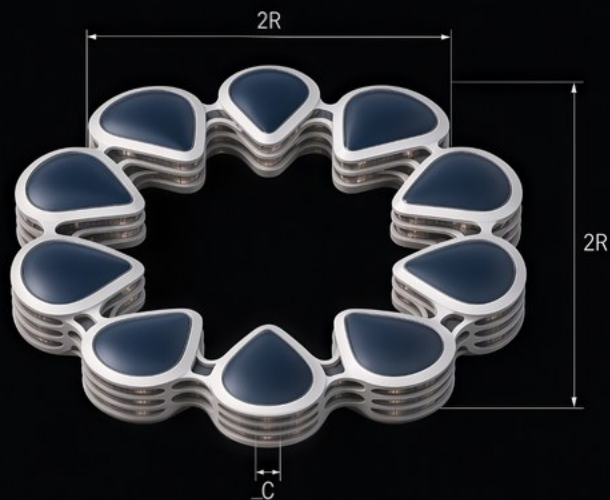


## A. GOVERNING GEOMETRY (PARAMETRIC CONTROL PACKAGE)

A. الهندسة الحاكمة (حزمة التحكم البارامتري)

### المنطق الهندسي

- وحدة أساسية حلقيّة متكررة
- عدد الوحدات N (10)
- نصف قطر الحلقة R
- شكل الوحدة (دمعة)
- ارتفاع الميني Z
- سماكة الحلقة T
- الفتحات البينية C



### GEOMETRIC LOGIC

- 01 Create a ring of radius R
- 02 Divide into N equal segments
- 03 Place modules tangentially
- 04 Connect with smooth bridges
- 05 Apply thickness T and set height Z
- 06 Ensure interstitial opening C

|        |                          |            |                                                             |
|--------|--------------------------|------------|-------------------------------------------------------------|
| N      | 10                       | R, Z, T, C | Governed variables                                          |
| Levels | Three — identical height | Style      | Modern, high-end, high-contrast — black, dark gray, silver. |

46,000 M<sup>2</sup> TOTAL GFA

|                                                                                                                                            |           |
|--------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Ten Units (10 × 3,150 m²)                                                                                                                  | 31,500 m² |
| Connectors (30 × 40 m²)                                                                                                                    | 1,200 m²  |
| Hall and Administration Pavilion<br>800-seat hall, lobbies, admin, hospitality, prayer, clinic, visitor services                           | 5,500 m²  |
| Services and Utilities Building<br>Data center, SOC, NOC, BMS, UPS, cooling, substation, generators, batteries, water, security, logistics | 5,200 m²  |
| Shaded Walkways and Arcades                                                                                                                | 2,600 m²  |
| TOTAL SHARED                                                                                                                               | 14,500 M² |

- Central oasis: open, unbuilt, not counted in GFA
- No structure or bridge across the oasis
- No basement below sea level
- Service trenches only
- All service and utility buildings located outside the ring
- Raised service platforms above flood datum



مركز الشيخ مشعل الأحمد الجابر الصباح  
للشباب والتكامل السيادي الوطني

# HHMYC

DESIGN BASIS REGISTER

Rev 1.0 | September 2026

Sheikh Mishal Al-Ahmad Al-Jaber Al-Sabah Center  
for Youth and National Sovereign Integration (HHMC)

*A Pledge and a Promise... For the Future of Kuwait.*

STATUS: GOVERNING REGISTER

OVERALL SITE PLAN

Master plan, program, conceptual package, RFP, QTO, Bidding

Master plan, program, conceptual package, RFP, QTO, Bidding

Where a document conflicts, this register prevails.

Hall, services and walkways sit outside the ring. Oasis remains open.



Sequence from the south entry: 10 → 8 → 9 → 4 → 5 → 3 → 2 → 1 → 7 → 6. Select a unit on the ring.

10

Units

3,470

Concurrent capacity in units

5,000 – 5,500

Site peak capacity

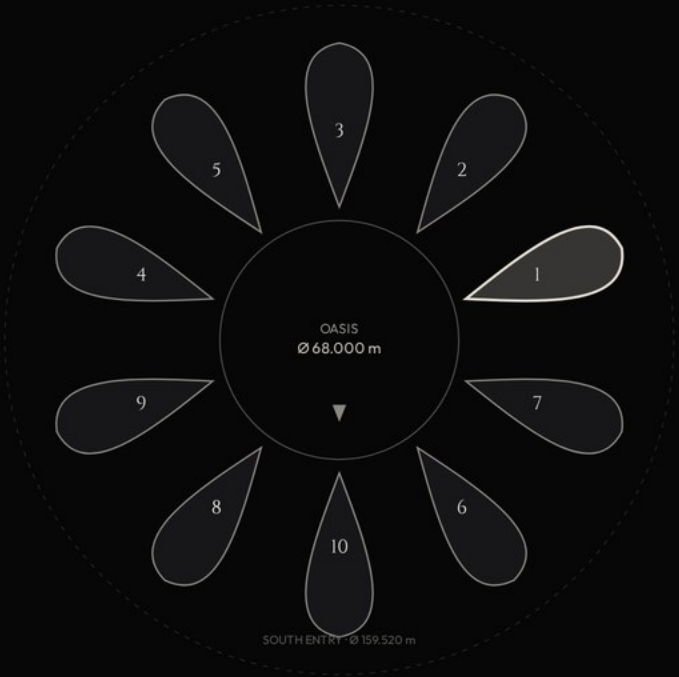
LOCAL DOUBLE HEIGHT

Studio height (Unit 8)

UNIT 01

RECEPTION & ORIENTATION

Arrival, wayfinding, visitor services



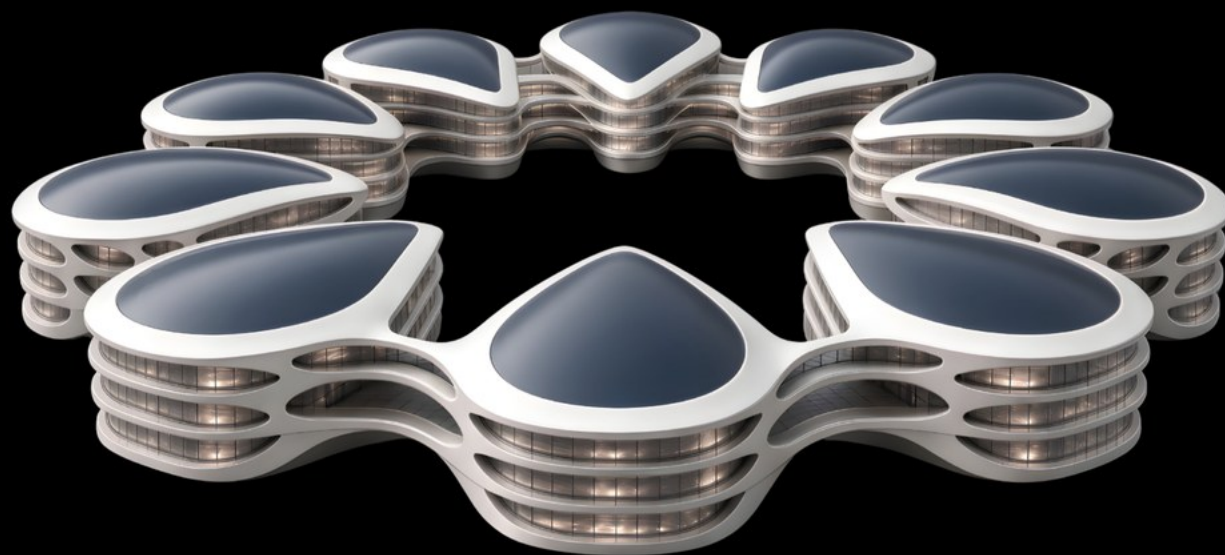
|                                                                                                                                            |                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary structure</b><br><br>Coated carbon steel A992 / Q355 equivalent; composite deck floors; braced cores; radial axes R0–R7 at 6 m. | <b>Roof gridshell</b><br><br>CHS members with cast or laser-cut nodes on a ring beam at +13.6 m; ground-assembled, single lift per unit.   |
| <b>BIPV glazing</b><br><br>Laminated 8 + PV + 8 mm (max ~20 mm total); dark navy solar surface.                                            | <b>Facade envelope</b><br><br>GFRC panels + double or triple insulating laminated glass with anti-glare coating.                           |
| <b>Exposed marine steel</b><br><br>Stainless SS 316L for handrails, external stairs, spray-zone details only — not the primary frame.      | <b>Foundations</b><br><br>Piled raft or pad-and-strip, to be determined by geotechnical investigation. No pile length or count is adopted. |
| <b>Repetition</b><br><br>One master kit × 10. Prototype unit 01 complete before serial production. All kits in 40-foot containers.         | <b>Steel tonnage (benchmark)</b><br><br>2,600 – 3,700 t. Benchmark only; manufacturer ROM governs.                                         |

Shipping: all kits in 40-foot containers; connections disassembled; marine delivery. One master kit × 10 — prototype unit 01 complete before serial.

☀ BIPV integrated roofs with high-efficiency photovoltaic glass — dark navy, not decorative.

🌊 Direct seawater intake from the open sea beyond the breakwater. The lagoon is a microclimate, not a cooling source.

🛡 Sovereign data infrastructure above flood datum. Specification-based, vendor-neutral in security scope.



Each colour marks a construction / systems zone. Materials and grades below are the Rev 1.1 basis. Where this panel and the Register would conflict, the Register prevails.

|                                    |                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------|
| ■ Primary structural frame         | Coated carbon steel A992 or Q355 equivalent; composite deck floors; braced cores; radial axes R0–R7 at 6 m  |
| ■ BIPV roof glazing                | Laminated 8 + PV + 8 mm (max ~20 mm); dark navy solar surface                                               |
| ■ Façade envelope                  | GFRC panels + double or triple insulating laminated glass with anti-glare coating                           |
| ■ Sovereign digital infrastructure | Data center design target Tier IV or III; SOC, NOC, BMS; vendor-neutral selection at RFQ                    |
| ■ Foundations                      | Piled raft or pad-and-strip to be determined by geotechnical investigation; no pile length or count adopted |
| ■ Ornamental details               | Islamic ten-point star geometry; discreet gold accents in interior finishes only                            |

|                        |                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------|
| ■ Roof gridshell       | CHS members with cast or laser-cut nodes on a ring beam at +13.6 m; ground-assembled, single lift per unit           |
| ■ Exposed marine steel | SS 316L for handrails, external stairs, spray-zone details only (not primary frame)                                  |
| ■ Cooling              | District cooling; seawater intake from the open sea beyond the breakwater; titanium plate heat exchangers            |
| ■ Utilities            | HDPE / GRP for water and drainage; standard cable trays; duplex fittings only at seawater heat-exchanger connections |
| ■ Sustainability       | LEED target + Kuwait MEW R-6 mandatory; grey-water reuse; open oasis and lagoon as passive microclimate              |

In sovereign, security, network, and data-center scopes, selection is specification-based and vendor-neutral, decided at the formal RFQ stage.

| SCOPE                                | NAMED / BENCHMARK                                        | STATUS | PACKAGE OWNER              | NOTES                                          |
|--------------------------------------|----------------------------------------------------------|--------|----------------------------|------------------------------------------------|
| Primary structural steel             | Baosteel (BSN, PRC)                                      | OK     | Main Works Contractor      | High-strength steel, mill certs, traceability. |
| Roof gridshell + diagrid             | Zhejiang Southeast Space Frame (PRC)                     | OK     | Main Works Contractor      | Parametric fabrication, hot-dip galvanizing.   |
| Third-source quality benchmark       | Hyundai Steel (KOR)                                      | OK     | Main Works Contractor      | Benchmark and alternate supply assurance.      |
| Façade GFRC panels                   | Tender · Rieder, Fibrobeton                              | INFO   | Façade Subcontractor       | GFRC panels with anchorage system.             |
| Insulating and laminated glass       | Tender · Guardian, AGC, Saint-Gobain                     | INFO   | Façade Subcontractor       | IGU with low-E, laminated safety glass.        |
| BIPV roof panels                     | Tender · Onyx Solar, LONGi BIPV, ML System               | INFO   | MEP / Renewable Contractor | Integrated PV with waterproofing system.       |
| District cooling equipment           | Tender · titanium exchangers per E-Section               | INFO   | MEP Contractor             | Chillers, titanium heat exchangers, pumps.     |
| Data center Tier IV / III (target)   | Vendor-neutral specification                             | OPEN   | Data Center Contractor     | Uptime Institute aligned. Named only at RFQ.   |
| Networking (LAN / WAN / backbone)    | Vendor-neutral. Huawei / Nokia sovereign scope: excluded | REJ    | ICT / Network Contractor   | Sovereign-compliant equipment and software.    |
| SOC / cybersecurity                  | Vendor-neutral. Kaspersky: excluded (US-sanctioned)      | REJ    | Cybersecurity Contractor   | 24/7 SOC, SIEM, threat intelligence.           |
| Fire protection                      | Tender · Honeywell, Tyco, Siemens                        | INFO   | Fire Protection Contractor | Sprinklers, gas suppression, FM-Approved.      |
| Elevators                            | Tender · KONE, Mitsubishi, Schindler                     | INFO   | Lift Contractor            | High-speed lifts, destination control.         |
| AV, display walls, interactive       | Tender · Samsung, LG, Barco, disguise                    | INFO   | AV / Media Contractor      | LED walls, projection, interactive systems.    |
| Water treatment                      | Tender · Veolia, SUEZ, ACWA                              | INFO   | Water Treatment Contractor | RO, UF, UV, grey-water reuse systems.          |
| Marine works (reclaimed islet, quay) | Kuwait / GCC tender — not in Chinese steel scope         | INFO   | Marine Contractor          | Reclamation, breakwater, dredging.             |

OK confirmed · INFO to be tendered · OPEN specification released / RFQ pending · REJ excluded / not compliant

Ranges where the input is uncertain. Any single number given without range or basis has been removed.

| Metric                       | Value                       | Basis                                             |
|------------------------------|-----------------------------|---------------------------------------------------|
| BIPV roof area (10 units)    | ~9,000 m²                   | Ten low double-curved shells, 38 × 28 m each      |
| Projected BIPV peak capacity | 1.2 – 1.5 MWp               | At 130–160 W/m² panel efficiency                  |
| Optional façade PV           | up to +0.5 MWp              | Vertical BIPV on selected façades                 |
| Coverage of consumption      | 20 – 35%                    | To be modelled post energy simulation             |
| Steel tonnage benchmark      | 2,600 – 3,700 t             | Benchmark only. Manufacturer ROM governs          |
| Roof shell rise-to-span      | 0.18                        | 6.8 m rise on 38 m span — not a pointed dome      |
| Building ring diameter       | 159.520 m                   | 10 identical units at 36°                         |
| Central open oasis           | Ø 68 m                      | No structure, no bridges, no service arc          |
| Data center design target    | Tier IV or III              | Design target, not certified before certification |
| Cooling intake               | Open sea, below thermocline | Not from the lagoon; titanium plate exchangers    |
| Water source                 | MEW network                 | No desalination. Grey-water reuse.                |
| Concurrent capacity in units | ~3,470 persons              | Per functional program                            |
| Site peak capacity           | ~5,000 – 5,500              | Subject to evacuation modelling                   |
| CapEx                        | Range per phase             | Open item H-6 in the Design Basis Register        |

PROHIBITED CLAIMS

- Structural efficiency multiplier “4×” vs baseline
- “20% material saving” without an FEA reference
- “Fatigue safety factor 8”
- “60–80% energy coverage” or “3.0–3.5 MW” output
- “Tier IV certified” before actual certification
- “Zero-Discharge” as a compliance claim
- Any pile length or count before geotechnical investigation

All values are concept-design indicators only and subject to change. Detailed engineering, simulation, and certification will refine final figures.

A. KUWAIT — MANDATORY

Kuwait Building Code (KBC) — as amended  
Mandatory for all construction and architectural works

Kuwait Fire Force (KFF) Life Safety Code  
Mandatory — fire, smoke and evacuation

MEW / R-6 (2014) Energy Conservation Code  
Mandatory — energy efficiency

Kuwait Environment Public Authority (KEPA) — EIA  
Mandatory — environmental assessment before construction

Kuwait Municipality coastal-strip regulations  
Mandatory — coastal height restriction; sovereign-project classification

Public Tenders Law — Law 49/2016  
Mandatory upon government tender

B. INTERNATIONAL — DESIGN REFERENCES

ISO 9001 / 14001 / 45001  
Quality, environment and safety

ISO 3834-2:2021  
Welding quality

AISC 360  
Steel design — reference

AWS D1.1  
Welding of steel structures

Eurocode 3 (EN 1993)  
Steel design — alternative reference

GB 50017  
Chinese steel structure code — fabrication

ASHRAE 90.1 / 62.1  
HVAC, energy efficiency and IAQ

NFPA 13, 72, 92  
Fire — design reference unless it conflicts with KFF

Uptime Institute Tier III / IV  
Data center resilience — design target

LEED v4 BD+C  
Sustainability — target certification

C. EXPLICITLY NOT APPLICABLE

Estidama Pearl Rating System (Abu Dhabi)  
Not applicable in Kuwait — replaced by LEED + MEW R-6

Aerospace alloy standards for architectural frame  
Al-Li 2195 rejected as a roof material

ORNAMENT

Islamic ten-point star geometry. Discreet gold accents in interior finishes only.

The ring is the emblem. There is no central dome, no service arc around the oasis, and no fourth floor. All ten units share one height, one kit, one outline.



|    |                                                             |
|----|-------------------------------------------------------------|
| 01 | SS 316L as primary structural frame                         |
| 02 | Al-Li 2195 aerospace alloy for the roof frame               |
| 03 | 50 mm glass panels                                          |
| 04 | Duplex SS 2205 for all piping (over-spec)                   |
| 05 | Central dome, central core, or service arc around the oasis |
| 06 | Fourth floor, graded unit heights, oculi                    |
| 07 | B1 annular basement or any structure below sea level        |

This catalogue is subordinate to the Design Basis Register. In case of any conflict, the Register prevails and this catalogue is updated — not the reverse.

|                 |                                                              |
|-----------------|--------------------------------------------------------------|
| Document        | HHMYC Design Basis Register · Technical Specifications Panel |
| Revision        | Rev 1.0 · Panel Rev 1.1                                      |
| Date            | September 2026                                               |
| Governing basis | HHMC Design Basis Register — this register prevails          |
| Status          | Concept design — not for construction                        |
| Confidentiality | Confidential — for evaluation only                           |
| Prepared by     | Eng. Hamad Alotaibi                                          |
| Role            | Strategic Advisor · Founder · Sovereign Systems Engineer     |
| Issuer          | Cohesion Holding · Kuwait                                    |
| Copyright       | Cohesion Holding — All rights reserved 2026                  |

ISSUED BY

ENG. HAMAD ALOTAIBI

Strategic Advisor · Founder · Sovereign Systems Engineer

Cohesion Holding

ALL RIGHTS RESERVED 2026 · KUWAIT

✉ cohesionkw@gmail.com

☎ +965 51179990



مركز الشيخ مشعل الأحمد الجابر الصباح للشباب والتكامل السيادي الوطني

Sheikh Mishal Al-Ahmad Al-Jaber Al-Sabah Center for Youth and National  
Sovereign Integration

# HHMYC

DESIGN BASIS REGISTER

A PLEDGE AND A PROMISE... FOR THE FUTURE OF KUWAIT



Governing register



Parametric certainty



Integrated design



National sovereignty

ENG. HAMAD ALOTAIBI

Strategic Advisor · Founder · Sovereign Systems Engineer

Cohesion Holding · All rights reserved 2026 · Kuwait

cohesionkw@gmail.com

+965 51179990